NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - IV

Estimation and Testing of Hypothesis, Large Sample Tests

Syllabus: Estimation - parameters, statistics, sampling distribution, point estimation, formulation of null and alternative hypothesis, critical and acceptance regions, level of significance, two types of errors and power of the test. Large sample tests: single proportion, difference of proportions, single mean, difference of means. Confidence interval for parameters in one sample and two sample problems.

I. Basic Terms

- **Population (universe)** : the complete collection of items under study. **Size N**.
- **Sample** : a part of the population selected for study. **Size n**. ($n \geq 30$: large sample, $n < 30$: small sample)
- **Parameter** : a numerical characteristic of the population, e.g. mean μ, variance σ^2, proportion P.
- **Statistic** : a numerical characteristic of the sample, e.g. $\bar{x}$, s^2, p. A statistic is a random variable.
- **Sampling** : random sampling with replacement (SRSWR) or without replacement (SRSWOR).

Sampling distribution : the probability distribution of a statistic computed from all possible samples of the same size n drawn from a population.

Standard error (S.E.) : standard deviation of the sampling distribution of a statistic. It measures the reliability of the statistic.

Important results

Statistic	Mean	Standard error
Sample mean $\bar{x}$	μ	$\sigma/\sqrt{n}$
Sample proportion p	P	$\sqrt{(PQ/n)}$
Difference of means $\bar{x}_1 - \bar{x}_2$	$\mu_1 - \mu_2$	$\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
Difference of proportions $p_1 - p_2$	$P_1 - P_2$	$\sqrt{(P_1Q_1/n_1 + P_2Q_2/n_2)}$

($Q = 1 - P$). If population is finite and sampling is without replacement multiply S.E. by $\sqrt{[(N - n)/(N - 1)]}$.

Central Limit Theorem (CLT)

If a random sample of size n is drawn from **any** population with mean μ and variance σ^2, then for large n ($n \geq 30$) the sampling distribution of $\bar{x}$ is approximately **normal** $N(\mu, \sigma^2/n)$. Hence $\mathbf{Z = (\bar{x} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)}$.

2. Estimation

Estimation is the process of finding the value of a population parameter using sample data. The rule/formula used is called an

estimator, its value for a given sample is an **estimate**.

2.1 Point estimation

A **single value** is given as estimate of the parameter. Example : $\bar{x}$ is a point estimate of μ ; s^2 estimates σ^2 ; p estimates P.

Characteristics of a good estimator

- **Unbiased** : $E(T) = \theta$ (e.g. $E(\bar{x}) = \mu$). Sample variance with divisor $(n - 1)$ is unbiased for σ^2.
- **Consistent** : $T \to \theta$ as $n \to \infty$ (variance of T tends to 0).
- **Efficient** : among unbiased estimators, the one with minimum variance.
- **Sufficient** : T uses all information about θ contained in the sample.

2.2 Interval estimation

An interval (L, U) such that $P(L < \theta < U) = 1 - \alpha$. It is called a **$(1 - \alpha) \times 100$ % confidence interval**; L and U are **confidence limits**. Example : 95 % C.I. means 95 times out of 100 the interval contains the true parameter.

Confidence level	90 %	95 %	98 %	99 %
Critical value z	1.645	1.96	2.33	2.58

3. Testing of Hypothesis : Concepts

A **statistical hypothesis** is a statement about a parameter / distribution of the population which is tested on the basis of

sample.

3.1 Null and alternative hypothesis

- **Null hypothesis H_0** : hypothesis of "no difference", tested for rejection. Example $H_0 : \mu = \mu_0$.
- **Alternative hypothesis H_1** : what we accept if H_0 is rejected. Examples $H_1 : \mu \neq \mu_0$ (two-tailed), $H_1 : \mu > \mu_0$ (right-tailed), $H_1 : \mu < \mu_0$ (left-tailed).

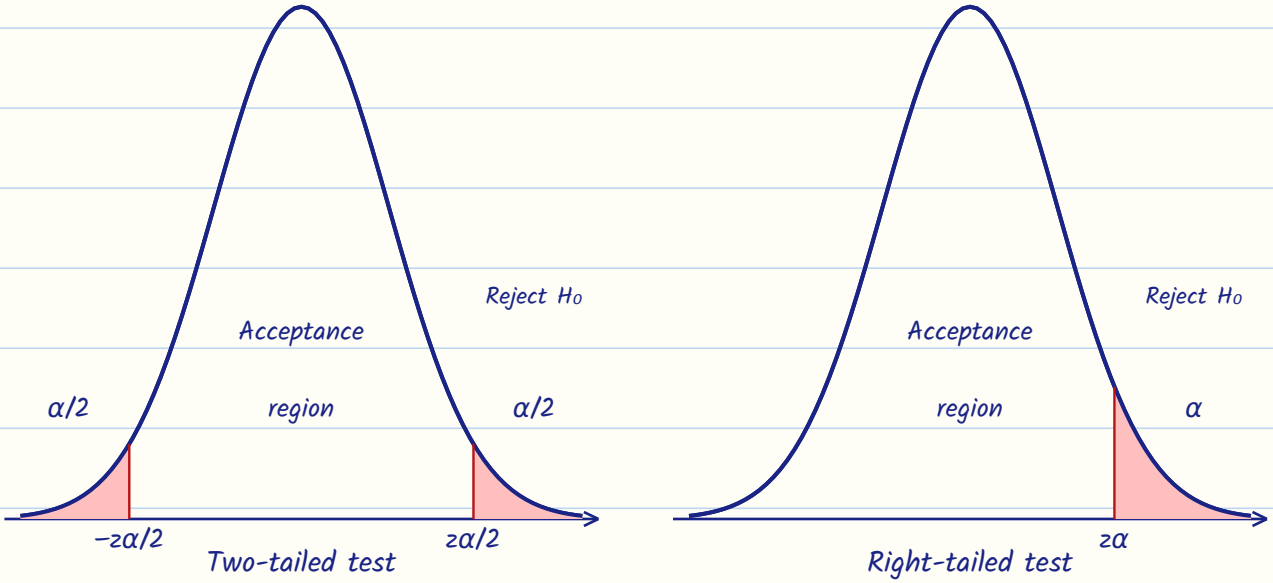
3.2 Test statistic, critical region

- **Test statistic** : a function of the sample values used to decide (like z).
- **Critical (rejection) region** : set of values of test statistic for which H_0 is rejected.
- **Acceptance region** : the remaining values where H_0 is accepted.
- **Critical value** : the boundary between the two regions.

3.3 Level of significance (α)

Probability of rejecting H_0 when it is true. Usually **5 %** or **1 %** (sometimes 10 %).

Level of significance	1 %	5 %	10 %						
Two-tailed test		z	= 2.58		z	= 1.96		z	= 1.645
Right-tailed test	z = 2.33	z = 1.645	z = 1.28						
Left-tailed test	z = −2.33	z = −1.645	z = −1.28						



3.4 Two types of errors

Decision / Truth	H_0 true	H_0 false
Reject H_0	**Type I error** (α)	Correct (power $1-\beta$)
Accept H_0	Correct	**Type II error** (β)

Type I error : rejecting H_0 when it is true, $P = \alpha$ (producer's risk). **Type II error** : accepting H_0 when it is false, $P = \beta$ (consumer's risk). **Power of the test** $= 1 - \beta$ = probability of rejecting H_0 when H_1 is true.

For a fixed sample size, decreasing α increases β. Increasing n reduces both.

3.5 Procedure of testing

- 1. Set up H_0 and H_1 (decides one or two-tailed).
- 2. Choose the level of significance α.
- 3. Select the suitable test statistic and compute its value.
- 4. Find the critical value from the table.
- 5. Decision : if |computed| > critical value, **reject** H_0, otherwise

accept H_0 (no evidence against it).

- 6. Write the conclusion in words.

4. Large Sample Tests ($n \geq 30$) : Z-tests

By CLT the sampling distributions are normal, so the test statistic

Z = (statistic − E(statistic)) / S.E.(statistic) ~ $N(0, 1)$.

4.1 Test for single mean

$H_0 : \mu = \mu_0$. \$ $Z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ (if σ unknown use s in place of σ)

Example 1

A sample of 400 items has mean 99. Population mean claimed is 100 with S.D. 8. Test at 5 % level.

$H_0 : \mu = 100$; $H_1 : \mu \neq 100$ (two-tailed).

$Z = (99 - 100)/(8/\sqrt{400}) = -1/0.4 =$ **−2.5**

$|Z| = 2.5 > 1.96$, so H_0 is **rejected** at 5 % level. (At 1 % level the critical value is 2.58, so H_0 is accepted at 1 %.)

4.2 Test for single proportion

Sample proportion $p = x/n$, $H_0 : P = P_0$. \$ $Z = (p - P_0) / \sqrt{(P_0Q_0/n)}$

Example 2

In a sample of 400 articles, 30 are defective. Can we say the proportion of defectives in the lot is 5 % ? ($\alpha = 0.05$)

$H_0 : P = 0.05$; $H_1 : P \neq 0.05$. $p = 30/400 = 0.075$.

S.E. = $\sqrt{(0.05 \times 0.95/400)} = 0.01090$

$Z = (0.075 - 0.05)/0.01090 = \mathbf{2.29} > 1.96 \rightarrow H_0$ **rejected** : proportion of defectives is not 5 %. (At 1 % level 2.29 < 2.58, accepted.)

4.3 Test for difference of means

$H_0 : \mu_1 = \mu_2$. $$ $Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$

Example 3

Two samples of 1000 men each from two places have mean heights 67.5 and 68.0 inches with S.D. 2.5 and 2.8. Test whether the places differ in mean height (5 %).

$Z = (67.5 - 68.0)/\sqrt{(6.25/1000 + 7.84/1000)} = -0.5/0.1187 = \mathbf{-4.21}$

$|Z| = 4.21 > 1.96 \rightarrow H_0$ rejected. The mean heights differ **significantly**.

4.4 Test for difference of proportions

$H_0 : P_1 = P_2$. Use pooled proportion $\hat{p} = (x_1 + x_2)/(n_1 + n_2) = (n_1p_1 + n_2p_2)/(n_1 + n_2)$

$$Z = (p_1 - p_2) / \sqrt{[\ \hat{p}\ \hat{q}\ (1/n_1 + 1/n_2)\ ]}$$

Example 4

In a sample of 400 men of town A, 40 smoke ; in 600 men of town B, 30 smoke. Is there a difference in the proportion of smokers ? (5 %)

$p_1 = 40/400 = 0.10$, $p_2 = 30/600 = 0.05$, $\hat{p} = 70/1000 = 0.07$, $\hat{q} = 0.93$

S.E. = $\sqrt{[0.07 \times 0.93\ (1/400 + 1/600)]} = \sqrt{(0.0651 \times 0.004167)} =$

0.01647

$Z = 0.05/0.01647 = \mathbf{3.04} > 1.96 \rightarrow H_0$ rejected (difference is significant).

Summary of Z-tests

Test	H_0	Z statistic
Single mean	$\mu = \mu_0$	$(\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
Single proportion	$P = P_0$	$(p - P_0)/\sqrt{(P_0Q_0/n)}$
Two means	$\mu_1 = \mu_2$	$(\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
Two proportions	$P_1 = P_2$	$(p_1 - p_2)/\sqrt{(\hat{p}\hat{q}(1/n_1 + 1/n_2))}$

5. Confidence Intervals

5.1 One-sample problems

- Mean (σ known or n large) : $\bar{x} \pm z\ (\sigma/\sqrt{n})$
- Proportion : $p \pm z\ \sqrt{(pq/n)}$

Here z = 1.96 for 95 %, 2.58 for 99 %.

Example 5

A sample of 100 has mean 65, population S.D. = 5. Find the 95 % confidence interval for μ.

$\bar{x} \pm 1.96\ \sigma/\sqrt{n} = 65 \pm 1.96\ (5/10) = 65 \pm 0.98 \rightarrow$ **(64.02 , 65.98)**

For 99 % : $65 \pm 2.58(0.5) = (63.71 , 66.29)$. Higher confidence gives a wider interval.

Example 6 (proportion)

In a sample of 400 voters 240 favour a candidate. 95 % C.I. for the true proportion :

$p = 0.6$, S.E. $= \sqrt{(0.6 \times 0.4/400)} = 0.0245$. Interval $= 0.6 \pm 1.96(0.0245) = 0.6 \pm 0.048 \rightarrow$ **(0.552 , 0.648)**

5.2 Two-sample problems

- Difference of means : $(\bar{x}_1 - \bar{x}_2) \pm z\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
- Difference of proportions : $(p_1 - p_2) \pm z\sqrt{(p_1q_1/n_1 + p_2q_2/n_2)}$

Example 7

Using the data of Example 3, 95 % C.I. for $\mu_1 - \mu_2$:

$(67.5 - 68.0) \pm 1.96\ (0.1187) = -0.5 \pm 0.233 \rightarrow$ **(−0.733 , −0.267)**. Zero is not inside, which agrees with rejecting H_0.

Sample size for estimating the mean

For error not more than E with confidence $(1 - \alpha)$: $n = (z\sigma / E)^2$

Example : $\sigma = 5$, $E = 1$, 95 % $\rightarrow n = (1.96 \times 5/1)^2 = 96.04 \approx$ **97**.

5A. More Solved Problems

Example 8 : single mean, σ unknown (n ≥ 30)

A sample of 64 students has mean marks 52 and S.D. 8. Test whether the population mean is 50 (5 %).

$H_0 : \mu = 50$; $H_1 : \mu \neq 50$. $z = (52 - 50)/(8/\sqrt{64}) = 2/1 =$ **2.0**

$2.0 > 1.96 \rightarrow H_0$ rejected at 5 % (just significant). At 1 % level it

would be accepted.

Example 9 : one-tailed (left) test

A bulb maker claims average life 600 hours. A sample of 100 bulbs has mean 580 hours with S.D. 50. Is the claim valid or is the true life less ? (5 %)

$H_0 : \mu = 600$; $H_1 : \mu < 600$ (left-tailed). $Z = (580 - 600)/(50/10) = \mathbf{-4.0}$

Critical value for left tail at 5 % is −1.645. Since $-4.0 < -1.645$, H_0 **rejected** : life of bulbs is less than 600 hours.

Example 10 : one-tailed test for proportion

A medicine is said to cure 90 % patients. In a sample of 500, 440 cured ($p = 0.88$). Is the claim justified ? (1 %)

$H_0 : P = 0.90$; $H_1 : P < 0.90$. S.E. $= \sqrt{(0.9 \times 0.1/500)} = 0.01342$

$Z = (0.88 - 0.90)/0.01342 = \mathbf{-1.49}$. Critical value −2.33. Since $|-1.49| < 2.33 \rightarrow H_0$ **accepted** : claim can be accepted.

Example 11 : two means with sample S.D.

Sample 1 : $n_1 = 50$, $\bar{x}_1 = 82$, $s_1 = 6$. Sample 2 : $n_2 = 60$, $\bar{x}_2 = 80$, $s_2 = 7$. Test $H_0 : \mu_1 = \mu_2$ at 5 %.

$Z = (82 - 80)/\sqrt{(36/50 + 49/60)} = 2/\sqrt{(0.72 + 0.8167)} = 2/1.2397 = \mathbf{1.61}$

$1.61 < 1.96 \rightarrow H_0$ accepted : no significant difference.

Example 12 : confidence interval for difference of

From Example 4 : $p_1 = 0.10$ ($n_1 = 400$), $p_2 = 0.05$ ($n_2 = 600$).

S.E. $= \sqrt{[0.1(0.9)/400 + 0.05(0.95)/600]} = \sqrt{(0.000225 +}$

$0.0000792) = 0.01744$

95 % C.I. for $P_1 - P_2 = 0.05 \pm 1.96(0.01744) = 0.05 \pm 0.0342 \rightarrow$ **(0.016 , 0.084)**

Example 13 : Type I and Type II errors numerically

Test $H_0 : \mu = 50$ against $H_1 : \mu = 56$ using $n = 25$, $\sigma = 10$, rule : reject H_0 if $\bar{x} > 54$.

S.E. $= 10/\sqrt{25} = 2$.

$\alpha = P(\bar{x} > 54 \mid \mu = 50) = P(Z > (54 - 50)/2 = 2) = 1 - 0.9772$ $=$ **0.0228**

$\beta = P(\bar{x} \leq 54 \mid \mu = 56) = P(Z \leq (54 - 56)/2 = -1) =$ **0.1587**

Power $= 1 - \beta = 0.8413$.

Relation between CI and test

A two-sided test of $H_0 : \theta = \theta_0$ at level α rejects H_0 exactly when θ_0 lies **outside** the $(1 - \alpha)$ confidence interval. For Example 5, 66 is outside (64.02, 65.98), so $H_0 : \mu = 66$ would be rejected at 5 %.

Steps to avoid mistakes

- Check whether σ or only s is given ; for $n \geq 30$ both work with Z.
- Read the question for words like "less than", "greater than" (one-tailed) or "differ", "not equal" (two-tailed).
- For two proportions always use the **pooled** $\hat{p}$ in the test, but p_1q_1 and p_2q_2 in the confidence interval.
- Write conclusion in words : "there is a significant difference" or "no evidence against H_0".

p-value idea (extra)

p-value = probability of getting a result at least as extreme as observed, if H_0 is true. Small p-value ($< \alpha$) means reject H_0. For $Z = 2.5$ two-tailed $p = 2 \times 0.0062 = 0.0124$ which is < 0.05 : reject at 5 %.

6. Quick Revision

- Statistic vs parameter ; S.E. of $\bar{x} = \sigma/\sqrt{n}$; S.E. of $p = \sqrt{(PQ/n)}$.
- Type I error α, Type II error β, power $= 1 - \beta$.
- Critical z : 5 % two-tailed 1.96, one-tailed 1.645 ; 1 % two-tailed 2.58, one-tailed 2.33.
- Reject H_0 if $|Z|$ > critical value.
- C.I. : estimate $\pm$ z $\times$ S.E.

Exam questions

- Define null hypothesis, level of significance, type I and type II errors.
- A coin is tossed 400 times, head appears 216 times. Is the coin unbiased ? (z = 1.6, accept at 5 %).
- Test equality of means of two large samples (given n, $\bar{x}$, s).
- Find 95 % and 99 % confidence limits for the mean from a large sample.